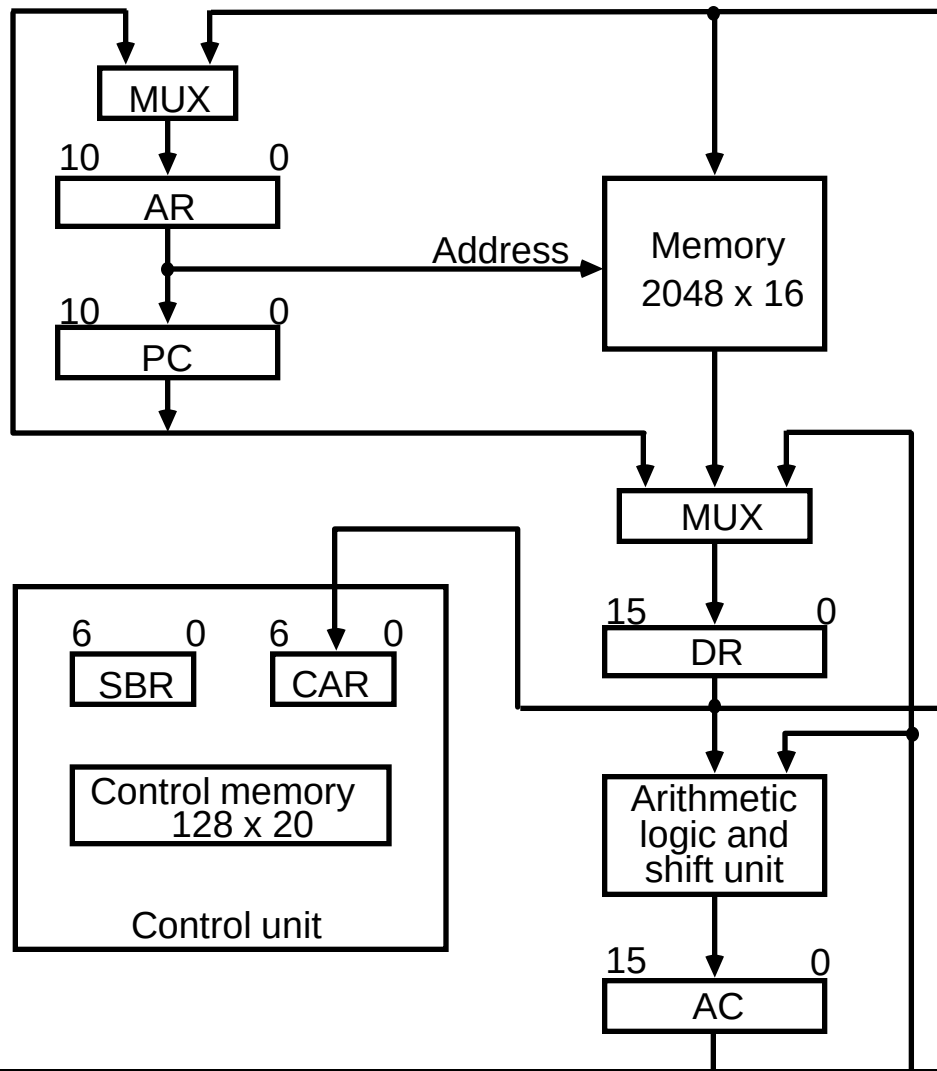


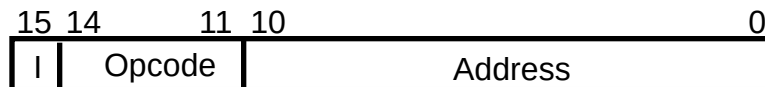
MICROPROGRAM EXAMPLE

Computer Configuration



MACHINE INSTRUCTION FORMAT

Machine instruction format

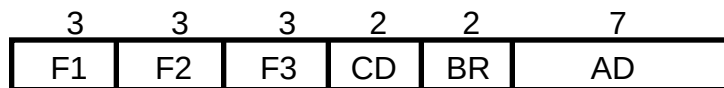


Sample machine instructions

Symbol	OP-code	Description
ADD	0000	$AC \leftarrow AC + M[EA]$
BRANCH	0001	if $(AC < 0)$ then $(PC \leftarrow EA)$
STORE	0010	$M[EA] \leftarrow AC$
EXCHANGE	0011	$AC \leftarrow M[EA], M[EA] \leftarrow AC$

EA is the effective address

Microinstruction Format



F1, F2, F3: Microoperation fields

CD: Condition for branching

BR: Branch field

AD: Address field



MICROINSTRUCTION FIELD DESCRIPTIONS - F1,F2,F3

F1	Microoperation	Symbol
000	None	NOP
001	$AC \leftarrow AC + DR$	ADD
010	$AC \leftarrow 0$	CLRAC
011	$AC \leftarrow AC + 1$	INAC
100	$AC \leftarrow DR$	DRTAC
101	$AR \leftarrow DR(0-10)$	DRTAR
110	$AR \leftarrow PC$	PCTAR
111	$M[AR] \leftarrow DR$	WRITE

F2	Microoperation	Symbol
000	None	NOP
001	$AC \leftarrow AC - DR$	SUB
010	$AC \leftarrow AC \vee DR$	OR
011	$AC \leftarrow AC \wedge DR$	AND
100	$DR \leftarrow M[AR]$	READ
101	$DR \leftarrow AC$	ACTDR
110	$DR \leftarrow DR + 1$	INCDR
111	$DR(0-10) \leftarrow PC$	PCTDR

F3	Microoperation	Symbol
000	None	NOP
001	$AC \leftarrow AC \oplus DR$	XOR
010	$AC \leftarrow AC'$	COM
011	$AC \leftarrow shl AC$	SHL
100	$AC \leftarrow shr AC$	SHR
101	$PC \leftarrow PC + 1$	INCPC
110	$PC \leftarrow AR$	ARTPC
111	Reserved	



MICROINSTRUCTION FIELD DESCRIPTIONS - CD, BR

CD	Condition	Symbol	Comments
00	Always = 1	U	Unconditional branch
01	DR(15)	I	Indirect address bit
10	AC(15)	S	Sign bit of AC
11	AC = 0	Z	Zero value in AC

BR	Symbol	Function
00	JMP	CAR $\leftarrow$ AD if condition = 1 CAR $\leftarrow$ CAR + 1 if condition = 0
01	CALL	CAR $\leftarrow$ AD, SBR $\leftarrow$ CAR + 1 if condition = 1 CAR $\leftarrow$ CAR + 1 if condition = 0
10	RET	CAR $\leftarrow$ SBR (Return from subroutine)
11	MAP	CAR(2-5) $\leftarrow$ DR(11-14), CAR(0,1,6) $\leftarrow$ 0

